

LAMP Stack — Error Types & Troubleshooting Guide

Common failure categories across Apache, MySQL, PHP and phpMyAdmin, with diagnosis and fixes

How This Guide Is Organized

Errors in a LAMP environment fall into **6 broad categories**. Each section lists the typical symptom, the most common cause, and the fix — in order of how often they occur in practice.

1. Apache (Web Server) Errors
2. MySQL / MariaDB (Database) Errors
3. PHP Errors
4. phpMyAdmin Errors
5. WordPress Errors
6. Permission & Ownership Errors
7. Network / Firewall / DNS Errors

1. Apache (Web Server) Errors

1.1 `Apache2 fails to start` / `Job for apache2.service failed`

- **Cause:** Port 80/443 already in use, or a syntax error in a config file.
- **Diagnose:**

```
sudo systemctl status apache2
sudo apache2ctl configtest
sudo journalctl -xeu apache2
```

- **Fix:**
 - If port is in use: `sudo lsof -i :80` then stop the conflicting service (often `nginx` or another web server).
 - If config syntax error: `apache2ctl configtest` shows the exact file/line; correct it and reload.

1.2 `403 Forbidden`

- **Cause:** Wrong file permissions, missing `index.html` / `index.php` , or `Require all denied` in the vhost.
- **Fix:**

```
sudo chown -R www-data:www-data /var/www/yoursite
sudo chmod -R 755 /var/www/yoursite
```

Check the `<Directory>` block has `Require all granted`.

1.3 404 Not Found

- **Cause:** Incorrect `DocumentRoot`, wrong URL path, or `.htaccess` rewrite rules misconfigured.
- **Fix:** Confirm `DocumentRoot` in the site config matches the actual file location; verify `mod_rewrite` is enabled if using clean URLs:

```
sudo a2enmod rewrite
sudo systemctl restart apache2
```

1.4 500 Internal Server Error

- **Cause:** PHP fatal error, bad `.htaccess` directive, or misconfigured module.
- **Diagnose:**

```
sudo tail -n 50 /var/log/apache2/error.log
```

- **Fix:** Read the exact error in the log (it names the file and line). Common culprits: missing PHP extension, syntax error in code, or an `.htaccess` directive not supported by the loaded modules.

1.5 AH00558: apache2: Could not reliably determine the server's fully qualified domain name

- **Cause:** Missing `ServerName` directive globally.
- **Fix:**

```
sudo "ServerName localhost" | sudo tee /etc/apache2/conf-
available/servername.conf
sudo a2enconf servername
sudo systemctl reload apache2
```

(This is a harmless warning, not a functional error, but it clutters logs.)

2. MySQL / MariaDB (Database) Errors

2.1 ERROR 2002 (HY000): Can't connect to local MySQL server through socket

- **Cause:** MySQL service isn't running, or the socket path is wrong.
- **Fix:**

```
sudo systemctl status mysql
sudo systemctl start mysql
```

If it won't start, check disk space (`df -h`) — MySQL refuses to start when the disk is full.

2.2 ERROR 1045 (28000): Access denied for user 'x'@'localhost'

- **Cause:** Wrong password, or the user/host combination doesn't exist in `mysql.user`.

- **Fix:**

```
sudo mysql
ALTER USER 'x'@'localhost' IDENTIFIED BY 'NewPassword123!';
FLUSH PRIVILEGES;
```

2.3 ERROR 1698 (28000): Access denied for user 'root'@'localhost'

- **Cause:** By default, Debian/Ubuntu MySQL configures `root` to use the `auth_socket` plugin (OS-level authentication), so a password login fails.
- **Fix:** Log in with `sudo mysql` (no password needed), then either switch root to password auth or, better, use a dedicated user (see Installation Guide, Step 3) instead of root.

2.4 Table doesn't exist / Unknown database

- **Cause:** Wrong database selected, or the database/table was never created/imported.
- **Fix:**

```
SHOW DATABASES;
USE your_database;
SHOW TABLES;
```

Re-import the `.sql` dump if the schema is genuinely missing:

```
mysql -u lampadmin -p your_database < backup.sql
```

2.5 MySQL Won't Start After a Crash / Corrupted InnoDB Tables

- **Diagnose:**

```
sudo tail -n 50 /var/log/mysql/error.log
```

- **Fix:** Temporarily enable `innodb_force_recovery` (start at level 1, increase only if needed) in `/etc/mysql/mysql.conf.d/mysqld.cnf`, restart, dump your data, then rebuild the instance.

3. PHP Errors

3.1 PHP code shows as plain text in the browser instead of executing

- **Cause:** `libapache2-mod-php` isn't installed/enabled, or Apache wasn't restarted after install.
- **Fix:**

```
sudo apt install libapache2-mod-php -y
sudo a2enmod php8.3 # match your installed version
sudo systemctl restart apache2
```

3.2 Fatal error: Uncaught Error: Call to undefined function mysqli_connect()

- **Cause:** The `php-mysql` (or `php-mysqli`) extension isn't installed.

- **Fix:**

```
sudo apt install php-mysql -y
sudo systemctl restart apache2
```

3.3 Blank White Screen (the "White Screen of Death")

- **Cause:** A fatal PHP error with `display_errors` turned off, so nothing is shown, but nothing is logged visibly either.
- **Fix:** Temporarily enable error display in `php.ini` for debugging:

```
display_errors = On
error_reporting = E_ALL
```

Then check:

```
sudo tail -n 50 /var/log/apache2/error.log
```

Always turn `display_errors` back Off in production.

3.4 PHP Fatal error: Allowed memory size exhausted

- **Cause:** Script exceeds `memory_limit` in `php.ini` (default is often 128M).
- **Fix:** Edit `/etc/php/8.3/apache2/php.ini` :

```
memory_limit = 256M
```

Then `sudo systemctl restart apache2` .

3.5 Class 'PDO' not found or missing extensions generally

- **Cause:** Required PHP extension not installed for that feature (PDO, GD, mbstring, etc.).
- **Fix:**

```
sudo apt install php-pdo php-gd php-mbstring -y
sudo systemctl restart apache2
```

4. phpMyAdmin Errors

4.1 404 Not Found when visiting `/phpmyadmin`

- **Cause:** Apache config for phpMyAdmin wasn't linked/enabled.
- **Fix:**

```
sudo ln -s /etc/phpmyadmin/apache.conf /etc/apache2/conf-
available/phpmyadmin.conf
sudo a2enconf phpmyadmin
sudo systemctl reload apache2
```

4.2 `mysqli_real_connect(): (HY000/1698): Access denied`

- **Cause:** Logging in with a user/password combination MySQL rejects (often trying `root` with a password when `auth_socket` is active).
- **Fix:** Log in to phpMyAdmin with a SQL user created with password authentication (e.g., `lampadmin`), not the system `root` account.

4.3 The configuration file now needs a secret passphrase (`blowfish_secret`)

- **Cause:** Missing or empty `$cfg['blowfish_secret']` in `config.inc.php`.
- **Fix:**

```
sudo nano /etc/phpmyadmin/config.inc.php
```

Set a random 32-character string:

```
$cfg['blowfish_secret'] = 'a-random-32-character-string-here!'
```

4.4 `Count(): Argument #1 ($value) must be of type Countable|array (PHP 8+ compatibility)`

- **Cause:** Older phpMyAdmin version incompatible with a newer PHP release.
- **Fix:** Update phpMyAdmin to a version that officially supports your PHP version, or downgrade PHP to a supported release.

4.5 Upload/Import Fails with Large SQL Files

- **Cause:** PHP's `upload_max_filesize` and `post_max_size` are too small.
- **Fix:** Edit `/etc/php/8.3/apache2/php.ini`:

```
upload_max_filesize = 64M
post_max_size = 64M
max_execution_time = 300
```

Restart Apache afterward.

5. WordPress Errors

5.1 Error establishing a database connection

- **Cause:** Wrong `DB_NAME`, `DB_USER`, `DB_PASSWORD`, or `DB_HOST` in `wp-config.php`, or MySQL isn't running.
- **Fix:**

```
sudo systemctl status mysql
```

Verify the credentials in `wp-config.php` match a working login:

```
mysql -u wp_user -p wordpress_db
```

If the database itself is corrupted, use **phpMyAdmin → Repair** on the affected tables, or:

```
REPAIR TABLE wp_options;
```

5.2 White Screen of Death on the WordPress Site

- **Cause:** A PHP fatal error, usually from a plugin/theme, or `memory_limit` exhausted.
- **Fix:** Enable debug mode temporarily in `wp-config.php` :

```
define( 'WP_DEBUG', true );  
define( 'WP_DEBUG_LOG', true );  
define( 'WP_DEBUG_DISPLAY', false );
```

Check `wp-content/debug.log` for the exact error. Disable the offending plugin by renaming its folder inside `wp-content/plugins/` via SSH/SFTP if the dashboard is inaccessible.

5.3 Error: The link you followed has expired (on plugin/theme upload)

- **Cause:** `upload_max_filesize` , `post_max_size` , or `max_execution_time` in `php.ini` are too low for the file being uploaded.
- **Fix:** Increase the limits (see phpMyAdmin section 4.5 for the same fix) and restart Apache.

5.4 Briefly unavailable for scheduled maintenance message stuck

- **Cause:** A failed/interrupted core, plugin, or theme update left a `.maintenance` file behind.
- **Fix:**

```
cd /var/www/mysite.local/public_html  
sudo rm .maintenance
```

5.5 Permalinks / Pages Return 404 Not Found

- **Cause:** `mod_rewrite` not enabled, or `AllowOverride` is `None` so the WordPress-generated `.htaccess` rules are ignored.
- **Fix:**

```
sudo a2enmod rewrite
```

Confirm `AllowOverride All` is set in the vhost's `<Directory>` block, then reload Apache and re-save permalinks in **Settings → Permalinks**.

5.6 Can't Upload Media / Install Plugins — Unable to create directory or Asks for FTP Credentials

- **Cause:** `wp-content` (and subfolders) aren't owned by `www-data` , so WordPress can't write directly and falls back to requesting FTP details.
- **Fix:**

```
sudo chown -R www-data:www-data /var/www/mysite.local/public_html/wp-content  
sudo find /var/www/mysite.local/public_html/wp-content -type d -exec chmod 755 {}  
sudo find /var/www/mysite.local/public_html/wp-content -type f -exec chmod 644 {}
```

```
\n
```

Also confirm `define( 'FS_METHOD', 'direct' );` is set in `wp-config.php`.

5.7 `Sorry, you are locked out` / Too Many Login Attempts (Admin Lockout)

- **Cause:** A security plugin blocked access after failed logins, or the admin password was lost.
- **Fix (reset password via database):**

```
UPDATE wp_users SET user_pass = MD5('NewPassword123!') WHERE user_login = 'admin';
```

Run this in phpMyAdmin or the MySQL CLI, then log in and immediately change the password from the dashboard.

5.8 Mixed Content Warnings After Enabling HTTPS

- **Cause:** `siteurl` / `home` in the database, or hardcoded links in content, still use `http://`.
- **Fix:** Update the site URLs:

```
UPDATE wp_options SET option_value = 'https://yourdomain.com' WHERE option_name IN ('siteurl','home');
```

Then use a search-and-replace plugin (or `wp search-replace` via WP-CLI) to fix `http://` references inside post content.

6. Permission & Ownership Errors

5.1 `Permission denied` writing to uploads/cache folders

- **Cause:** Apache runs as `www-data`, but the directory is owned by another user.
- **Fix:**

```
sudo chown -R www-data:www-data /var/www/yourwebsite/uploads
sudo chmod -R 775 /var/www/yourwebsite/uploads
```

5.2 `.htaccess` rules are ignored

- **Cause:** `AllowOverride None` set in the vhost, disabling `.htaccess` processing.
- **Fix:** In the site config, set:

```
<Directory /var/www/yourwebsite>
    AllowOverride All
</Directory>
```

Then `sudo systemctl reload apache2`.

7. Network / Firewall / DNS Errors

7.1 Site loads locally but not from outside the server

- **Cause:** Firewall (UFW/iptables) or cloud security group blocking port 80/443.
- **Fix:**

```
sudo ufw allow in "Apache Full"
sudo ufw status
```

Also check your cloud provider's security group / firewall rules if hosted on AWS, GCP, Azure, etc.

7.2 This site can't be reached (DNS)

- **Cause:** Domain not pointed at the server's IP, or DNS propagation still pending.
- **Fix:** Verify with:

```
dig yourdomain.com +short
nslookup yourdomain.com
```

Confirm the returned IP matches your server.

7.3 SSL certificate problem / mixed content warnings

- **Cause:** Certbot certificate expired, or the site loads some assets over `http://` instead of `https://`.
- **Fix:**

```
sudo certbot renew --dry-run
sudo certbot renew
```

Update hardcoded `http://` links in the codebase to `https://` or protocol-relative URLs.

Quick Reference — Where to Look First

Symptom	Log File to Check
Site won't load / crashes	<code>/var/log/apache2/error.log</code>
Apache access patterns	<code>/var/log/apache2/access.log</code>
MySQL connection/start issues	<code>/var/log/mysql/error.log</code>
PHP-specific fatal errors	Apache error log (PHP logs there by default)
WordPress-specific errors	<code>wp-content/debug.log</code> (after enabling <code>WP_DEBUG_LOG</code>)
System-level service failures	<code>sudo journalctl -xeu <service></code>

General Troubleshooting Rule: Always check the relevant log file first — the exact error message and line number will point directly to the cause in over 80% of cases.

